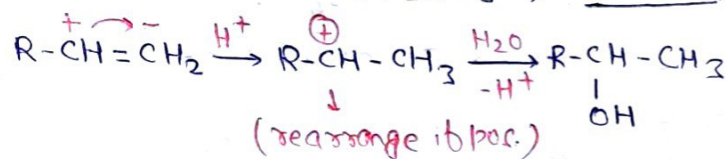


Alcohols, Phenols and Ethers

MOP of Alcohols

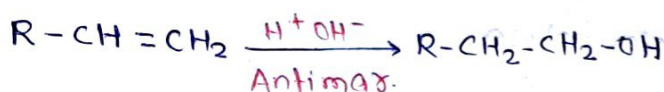
① From Alkenes

① With dil. Acid (conc. H_2SO_4/H_2O), or (H_2O/H_3O^+) (Markov).



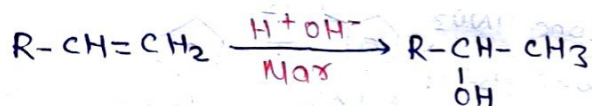
② H₂O (Hydroboration-oxidation)

Reagent: - (i) B_2H_6/THF
(ii) H_2O_2/OH^-



③ OMDM (Oxymercuration-Demercuration)

Reagent: - (i) $Hg(OAc)_2/H_2O$
(ii) $NaBH_4$



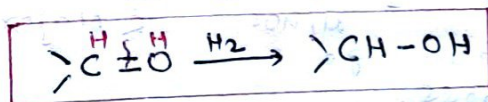
② Reduction Reaction

① By reduction of carbonyl compounds

Reagent: - ① LAH (Lithium aluminium hydride) or $LiAlH_4$ (source of H^-)

② SBH (Sodium borohydride) or $NaBH_4$ (source of H^-)

③ H_2 + catalyst



DM $\rightarrow$ $C=O$ not π , $CH-OH$ π bond KO, H, H set to do.

③ By reduction of carboxylic acid & Ester & Acid chloride

Mind maps

Carboxylic Acid

Ester

Acid chloride

① LAH

① LAH

① LAH

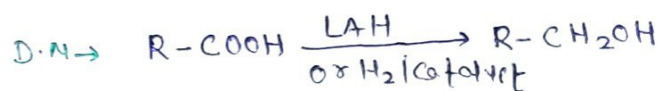
② H_2 + catalyst

② H_2 + catalyst

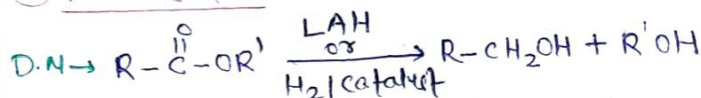
② H_2 + catalyst

③ SBH

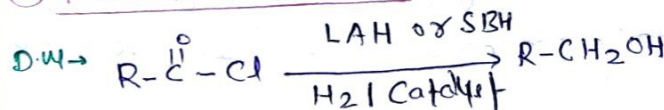
① From Carboxylic acid



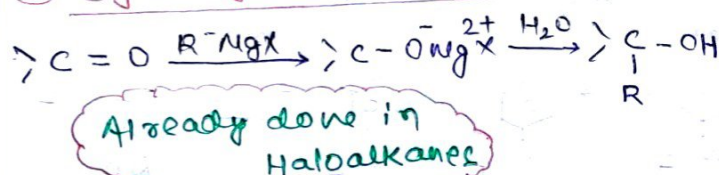
② From Ester



③ From Acid chloride

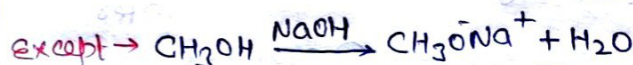
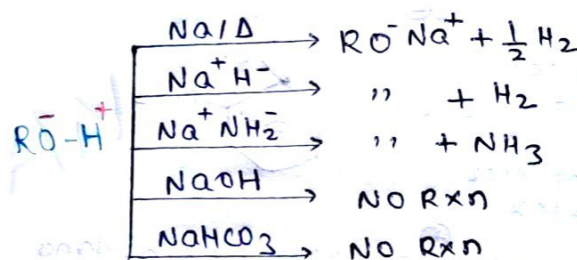


③ By Grignard reagent

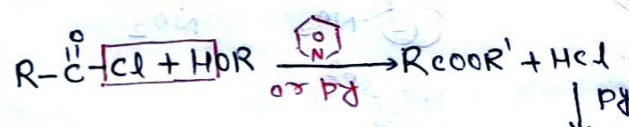
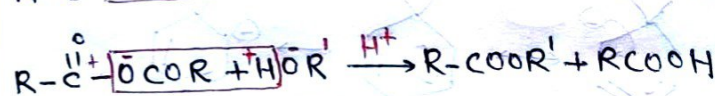
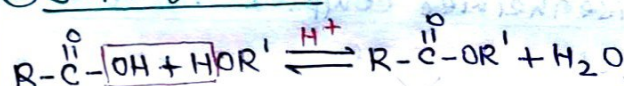


Properties of Alcohols

① Acidic Nature

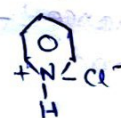


② Esterification



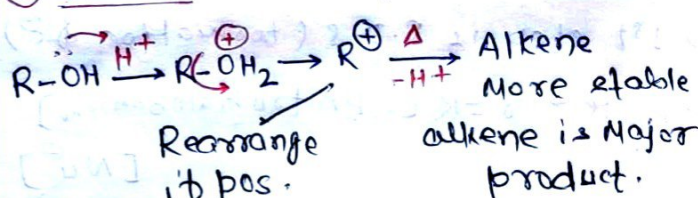
• $R-O-R$ $\propto$ stability of C^+ w.r.t. Acid

• $R-O-R$ $\propto \frac{1}{S.H} (1^\circ > 2^\circ > 3^\circ)$ w.r.t. Alcohol

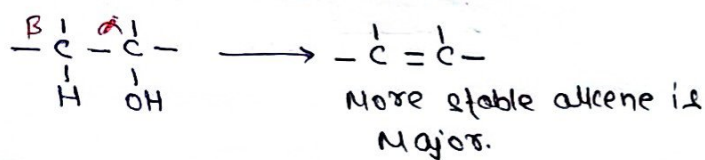


③ Dehydration of Alcohol

① Via E_1 (conc. H_2SO_4 or conc. H_3PO_4 , H^+/Δ)



⑥ Via E_2 (Anhy. $ZnCl_2$, P_2O_5/Δ , $POCl_3/\Delta$, Al_2O_3/Δ)
 NO rearrangement

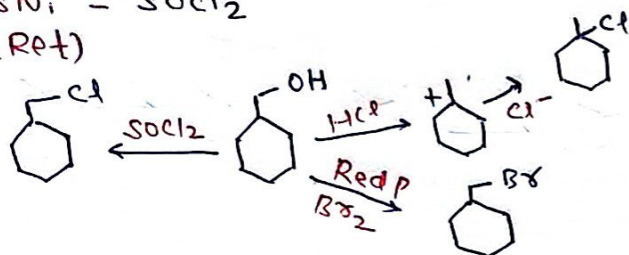


④ With HX , PX_3 (Red P + X_2), PX_5 , SOX_2

SN^1 - HX (Racemization)

SN^2 - PX_3 , PX_5 , $SOCl_2$ / py (Inv)

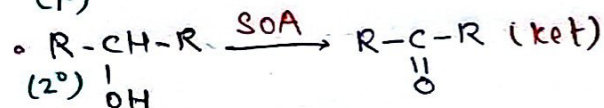
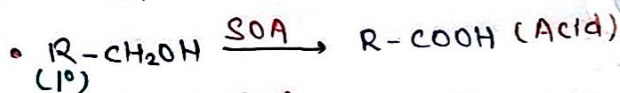
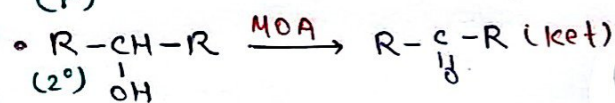
SN_i - $SOCl_2$ (Ret)



⑤ Oxidation of Alcohols

MOA $\rightarrow$ Mild oxidizing agent

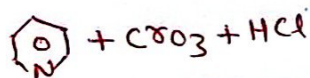
SOA $\rightarrow$ Strong oxidizing agent



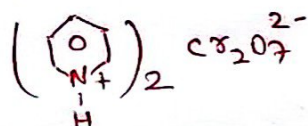
3° Alcohol $\xrightarrow[SOA]{MOA}$ No oxidation

Mild oxidizing Agents

① PCC (Pyridinium chlorochromate)



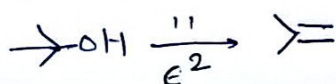
② PDC (Pyridinium dichromate)



③ CrO_3 + acetone or Anhy. CrO_3

④ $CrO_3 + CH_2Cl_2$ (Collins' reagent)
 or $2Py$ ($\text{Pyridine ring} \text{ } ^+$)

⑤ Cu or Ag / $300^\circ C$



Strong oxidizing Agents

① $KMnO_4 / H^+$ or Hot alk. $KMnO_4 / H^+$

② $K_2Cr_2O_7 / H^+$ or Hot "

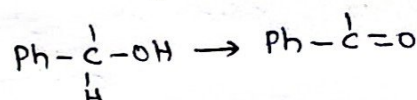
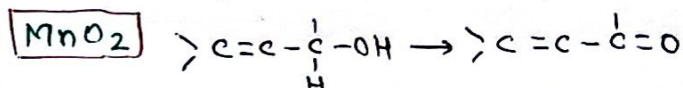
③ conc. HNO_3

④ Aq. CrO_3 or $CrO_3(aq)$ or $CrO_3 + H_2O$ or H_2CrO_4 .

⑤ $CrO_3 + H_2SO_4$ (Jones' reagent)

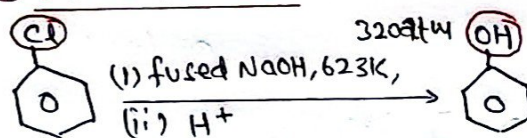
Special Mild oxidizing Agents

$\rightarrow$ For Allylic & Benzylic Alcohol only

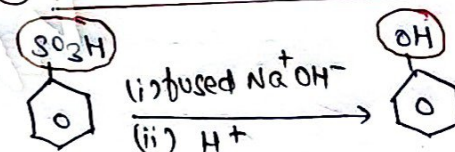


MOP of PHENOL

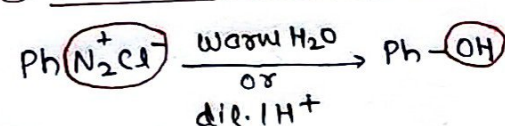
① Dow's Process



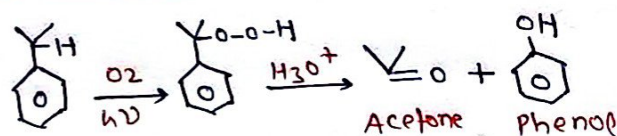
② From Benzene Sulphonic Acid



③ From Diazonium salt



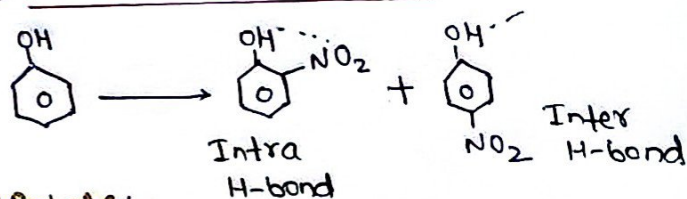
④ From Cumene



Properties of Phenol

1.E.S.R

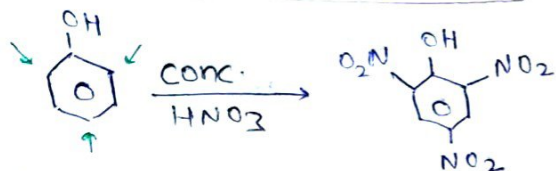
① with dilute nitric acid



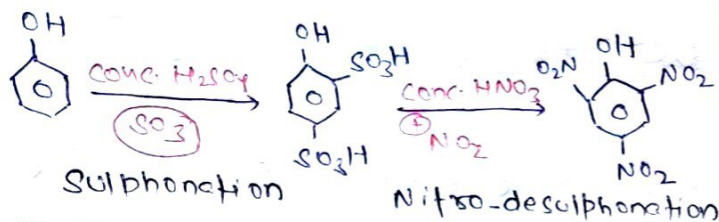
*Points:-

- B.P $\rightarrow$ (O < P) • Volatility $\rightarrow$ (O > P)
- Ortho & Para isomers can be separated by steam distillation.

② With Conc. Nitric Acid

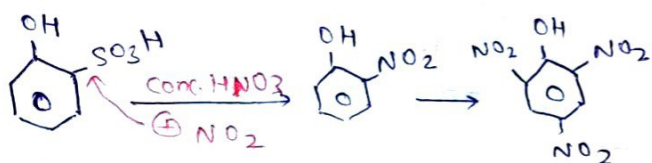


Nowadays

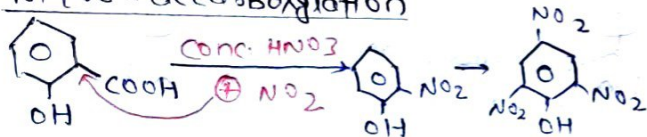


Special Rxn:-

(i) Nitro - Desulphonation

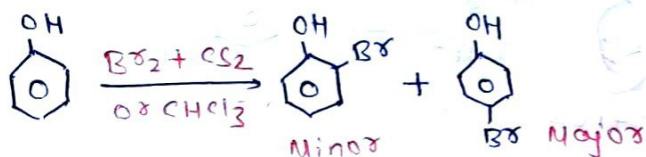


(ii) Nitro - decarboxylation

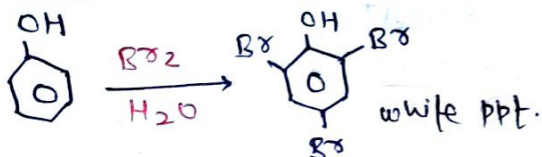


③ Mono bromination

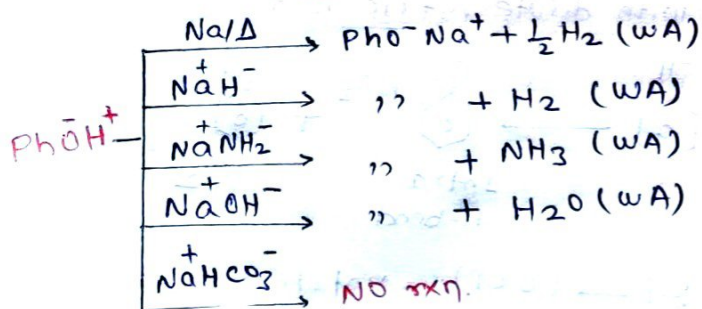
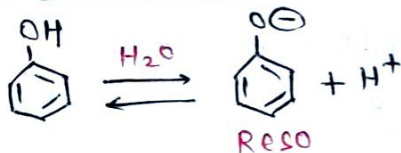
solvent $\rightarrow$ low polarity



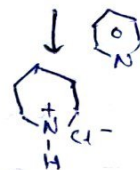
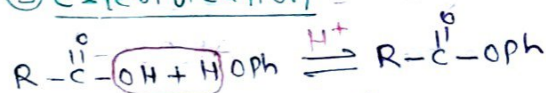
④ Tri bromination



② Acidic Nature

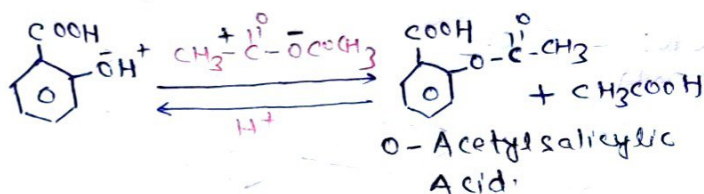


③ Esterification

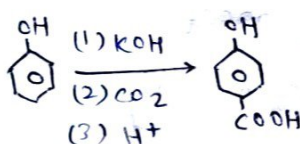
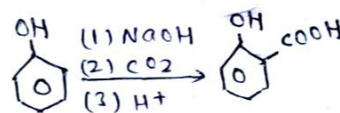


Aspirin formation

$\rightarrow$ Acetylation of Salicylic Acid



④ Kolbe's Rxn:-



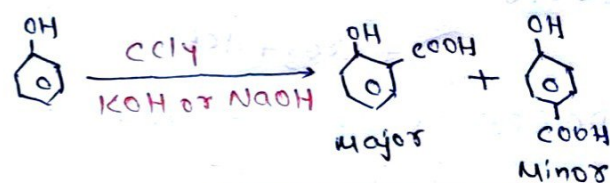
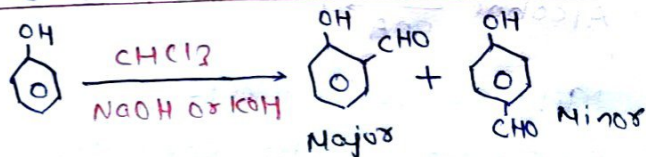
Charge density
 $\text{Na}^+ > \text{K}^+$
 $\downarrow$
ortho stable

⑤ Reimer - Tiemann Reaction

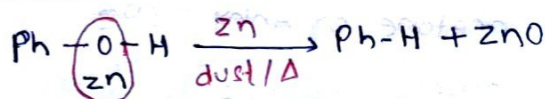
Reagent:- $\text{CHCl}_3 + \text{KOH}$ or NaOH

e^- or Attacking Reagent:-

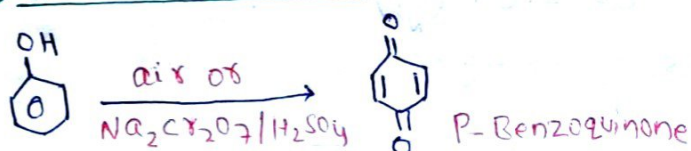
$\text{CCl}_2(\text{dCC}) \rightarrow$ Intermediate (dichloro carbene)



⑥ Reaction of Phenol with Zn/dust

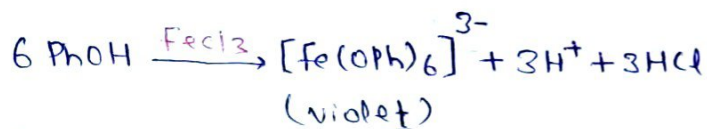


⑦ Oxidation of Phenol

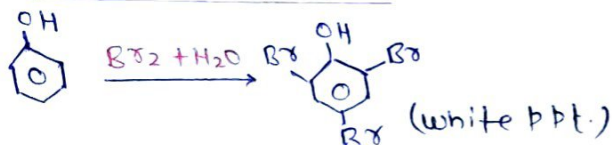


⑧ Test of Phenol

① FeCl₃ test

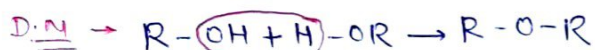
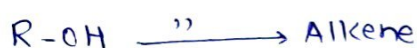
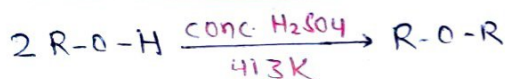


② Bromine water test

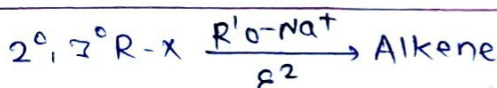
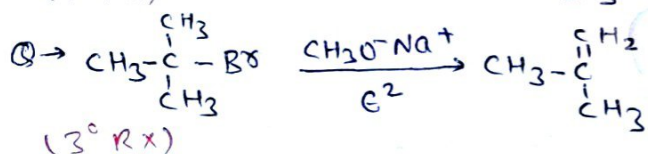
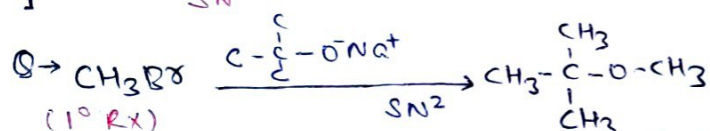
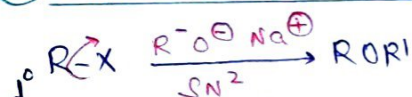


MOP of Ether

① Dehydration of Alcohols



② Williamson Ether Synthesis (S_N²)



following comp. donot give W.E.S rxn. (S_N²)

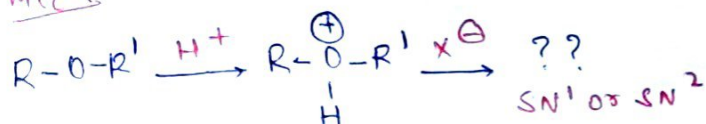
Comp.	Reasons
$\text{CH}_2=\text{CH-X}$	Partial double bond char.
Ph-X	"
	Bredt's rule

Properties of Ether

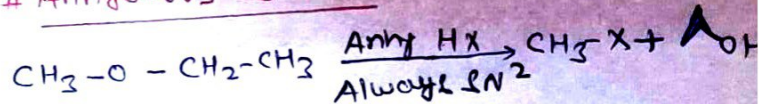
① Reaction with HX

- S_N2 → Normal 1°, 2°
- S_N1 → Normal 3°, Refu 1°, 2°, 3°
- In Anhydrous HX always S_N2.

Mech



Anhydrous cond.



Excess HX

- $\text{CH}_3\text{-O-CH}_2\text{-CH}_3 \xrightarrow[\text{HX}]{\text{Excess}} \text{CH}_3\text{X} + \text{CH}_3\text{CH}_2\text{X}$
- $\text{Ph-O-CH}_3 \xrightarrow{\text{''}} \text{PhOH} + \text{CH}_3\text{X}$
- $\text{Ph-CH}_2\text{-O-CH}_3 \xrightarrow{\text{''}} \text{PhCH}_2\text{X} + \text{CH}_3\text{X}$
- $\text{CH}_3\text{-C(CH}_3)_2\text{-O-CH}_3 \xrightarrow{\text{''}} \text{X} + \text{CH}_3\text{X}$

OP Point :-

↳ Vicinal di iodide are unstable due to very low B.E of C-I.

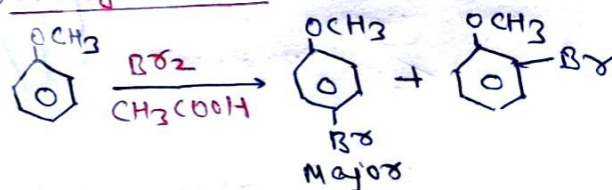


vic di iodide

② Electrophilic substitution rxn

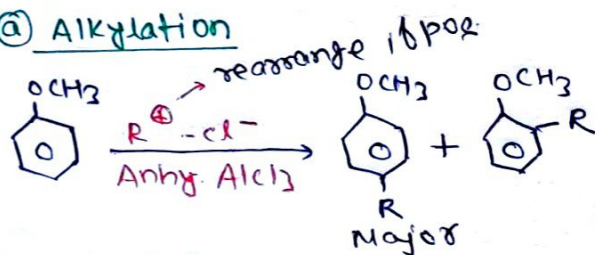
↳ Always para major

① Halogenation

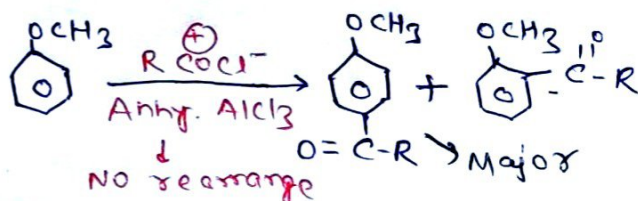


② Friedel crafts reactions

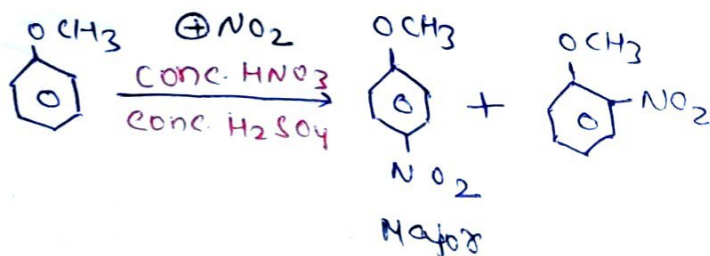
① Alkylation



② Acylation

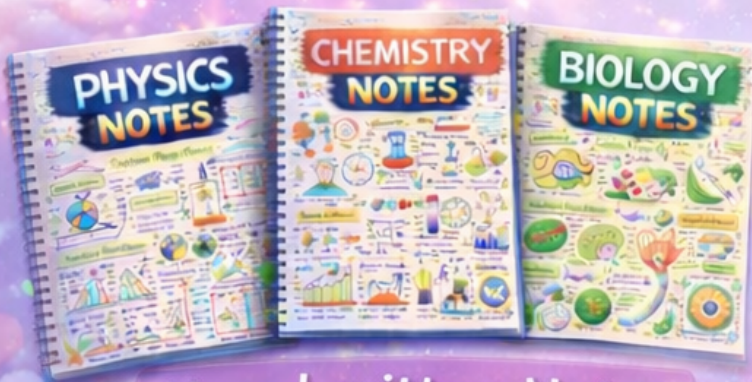


③ Nitration



By @THUNDERSTUDY

Other By @Thunderstudy



Handwritten Notes

By Pragma



Boards Dominator Series

By @Thunderstudy

Best Notes Best Book
For PCMB

Scan the QR or
click here to join >

